

Multiple Choice (10 marks)

1. Evaluate the determinant $\begin{vmatrix} 1 & 4 & 5 \\ \Delta & 123 & -31 \\ -2 & & \end{vmatrix}$
 - (a) -28
 - (b) 28
 - (c) -7
 - (d) 0
 - (e) 7
2. Find the magnitude of E for a plane wave in free space, if the magnitude of H for this wave is 5 amp/meter.
 - (a) 1200 volts/meter
 - (b) 5 ohms
 - (c) 1883 volts/meter
 - (d) 940.5 volts/meter
 - (e) 5 volts/meter
3. If a Hexadecimal number needs to convert to binary. For each hexadecimal digit, there will be how many bits
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) 8
4. Let $A = \begin{bmatrix} 1 & 12 \\ 283 & 323 \\ 151 & \end{bmatrix}$, find $A + B$ and $A - B$.
 - (a) $A + B = \begin{bmatrix} 473 & 375 \\ 375 & 475 \end{bmatrix}$, $A - B = \begin{bmatrix} -1 & -9 \\ -1 & 3 \end{bmatrix}$
 - (b) $A + B = \begin{bmatrix} 375 & 475 \\ 475 & 375 \end{bmatrix}$, $A - B = \begin{bmatrix} -2 & -9 \\ -1 & 2 \end{bmatrix}$
 - (c) $A + B = \begin{bmatrix} 473 & 475 \\ 475 & 375 \end{bmatrix}$, $A - B = \begin{bmatrix} -1 & -10 \\ -1 & 2 \end{bmatrix}$
 - (d) $A + B = \begin{bmatrix} 375 & 475 \\ 475 & 375 \end{bmatrix}$, $A - B = \begin{bmatrix} -2 & -9 \\ -2 & 1 \end{bmatrix}$
 - (e) $A + B = \begin{bmatrix} 375 & 474 \\ 474 & 375 \end{bmatrix}$, $A - B = \begin{bmatrix} -1 & -8 \\ -1 & 2 \end{bmatrix}$
5. A box which tells the effect of inputs on control sub system is known as
 - (a) Process box.
 - (b) State box.
 - (c) Input box.
 - (d) System box.
 - (e) Data Box.

True / False (10 marks)

Answer True or False

6. True or False: An electromagnetic wave of 40 KHz takes 0.7 milliseconds to travel 60 kilometers in a vacuum.
7. True or False: The energy stored in the magnetic field of the solenoid is 1.5 joule, which is significantly higher than the correct value.
8. True or False: Approximately 5 percent of the total volume of an iceberg floats above the water surface due to the density difference between ice and water.
9. True or False: The short circuit level at the busbar for four identical alternators rated at 20 MVA each is 600 MVA.
10. True or False: The magnetic intensity H in free space corresponding to the electric field E is given by $H = - \{iA / c_0\} \cos((t - (z/c)))$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the power expended in the field excitation of a shunt motor with a 110-ohm resistance and a 220-volt supply, and discuss the implications of this power on motor performance.
12. Analyze the power loss in a 5-mile transmission line made of two No. 0 copper wires carrying 75 amperes. Additionally, calculate the heat produced by an electric heater operating at 7.5 amperes and 110 volts.

End of Paper